

PYTHON 3 CHEAT SHEET

COLLECTION OF USEFUL NOTES FOR BEGINNERS

PYTHON IS AN HIGH-LEVEL GENERAL-PURPOSE PROGRAMMING LANGUAGE THAT IS QUICKLY GAINING POPULARITY. THIS IS A LIST WITH SOME OF THE CORE CONCEPTS AND VARIOUS IMPORTANT THINGS TO KEEP IN MIND WHILE STARTING OUT.



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Variables: **String**, **Integer**, **Float**, **Boolean**
str: "This here is a String", 'also this'
int: 1234, 0, -4321 (any whole number without decimals)
float: 9.95, 0.0, -2493.3 (any number with decimals)
boolean: True, False

Type (to check what kind of datatype):
variable = 1234
print(type(variable))

Escape Characters:
new line: \n
speech marks: \"
backslash: \\
tab: \t

Comments (for notes, won't execute):
#This is a single comment line.
"""to turn multiple lines into comments,
use triple quotes like this"""

Functions:
Block of code that does something more complex to avoid repetition.
- A Function is defined by using the def keyword
- Block of code which only runs when it is called
- Pros: less code, reduced code duplication, more descriptive
def my_func():
 print("This will print when func is called")
Call a function by using the function name followed by parenthesis.

List (stores a series of items):
my_list1 = ["Apple", "Banana", "Orange"]
my_list2 = [2, 22, 222, 2222]

Basic list operations:
- Create a list
- Add an item to the list = .append()
- Remove an item from the list = .remove()
- Change items in the list
- Access items of the list

Conditionals (if/elif/else):
- Very Important in coding generally. Set the boundaries, so to speak
- If statements run if required conditions are True, example below:

```
x = 3  
y = 6  
if y > x:  
    print("y is a larger number than x")  
elif y == x:  
    print("y and x have the same value")  
else:  
    print("x is a larger number than y")
```

Useful apps/libraries (there are tons more):
GUI: Kivy, PyQt, Tkinter, PySimpleGUI
Web: Django, Flask
Games: Pygame, Panda3D
Scientific: SciPy, Matplotlib, NumPy
Web Scraping: Scrapy, BeautifulSoup

TIP: Use comments only when needed, to guide your teammates or to remind yourself
TIP: "if" and "elif" do not need a "else" to execute properly
TIP: Conditions = logic, either true or false. USED most commonly in "if statements" and "loops"

Set (unordered, unchangeable, unindexed):
my_set1 = {"Apple", "Banana", "Orange"}
my_set2 = {2, 22, 222, 2222}

Input/Output Data:
Output: print("Hello World.")
Input: answer = input("Question")

Converting floats and Integers:
Convert float into integer:
int(9.95)
Convert integer into float:
float(1234)
Convert string into integer if string is a number:
int("999")

String Manipulation:
- Strings are arrays of characters
- You can get certain characters with square brackets:
variable = "example"
print(variable[4])

output: x (since it is the character number [1]. It always starts with [0]).
Type print(len(variable)) to get the number of characters in example = 7.
Type print(variable.upper()) to make the string uppercase = EXAMPLE
Type print(variable.lower()) to make the string lowercase = example
other examples: .strip to remove spaces,
.split to split a string into a list with a keyword such as " , "

Arithmetic operators:
Addition: +
Subtraction: -
Multiplication: *
Division: /
Modulus: %
Exponent: **
Floor division: //

Comparison operators:
Equals: a == b
Not equals: a != b
Less than: a < b
Less than or equal to: a <= b
Greater than: a > b
Greater than or equal to: a >= b
NOTE: = is not a comparison op.

TIP: Proper way of writing code is to have every piece of logic inside a function and having a proper description/name to each function.

While loops (execute a set of statements as long as a condition is true):

```
condition = 1  
while condition < 10:  
    print(condition)  
    condition += 1
```

The code above would run 9 times, until the limit of 10 is achieved.

while True:
 print("Basically runs forever")

The code above will create an infinite loop.

Project ideas	(beginner/intermediate):
Calculator	Password Manager
Quiz App	Reddit Bot
File Explorer	Music Player
Notepad Clone	Web Scraper
Speed Typing Test	Stock Visualizer